Fourier Series

complex transform

synthesis

$$x(t) = \sum_{k=-\infty}^{+\infty} X[k]\, e^{j\,2\pi kt/T}$$

analysis

$$X[k] = \frac{1}{T}\int_{0}^{T} x(t)\, e^{-j\,2\pi kt/T}\, dt$$

Time domain:
x(t) is complex, continuous and periodic
t runs over one period, from 0 to T

Frequency domain:
X[k] is complex, discrete, and aperiodic
k runs from negative to positive infinity
k > 0 are positive frequencies
k < 0 are negative frequencies

real transform

synthesis

$$x(t) = \sum_{k=0}^{+\infty} ReX[k]\cos(2\pi kt/T) - ImX[k]\sin(2\pi kt/T)$$

analysis

$$ReX[k] = \frac{2}{T}\int_{0}^{T} x(t)\cos(2\pi kt/T)\, dt$$

$$ImX[k] = \frac{-2}{T}\int_{0}^{T} x(t)\sin(2\pi kt/T)\, dt$$

Time domain:
x(t) is real, continuous, and periodic
t runs over one period, from 0 to T

Frequency domain:
Re X[k] is real, discrete and aperiodic
Im X[k] is real, discrete and aperiodic
k runs from zero to positive infinity

Note: Before using the synthesis equation, the value for Re X[0] must be divided by two.

Fourier Transform

complex transform

synthesis

$$x(t) = \int_{-\infty}^{+\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

analysis

$$X(\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} x(t)\, e^{-j\omega t}\, dt$$

Time domain:
x(t) is complex, continious and aperiodic
t runs from negative to positive infinity

Frequency domain:
X(ω) is complex, continious, and aperiodic
ω runs from negative to positive infinity
ω > 0 are positive frequencies
ω < 0 are negative frequencies

real transform

synthesis

$$x(t) = \int_{0}^{+\infty} ReX(\omega)\cos(\omega t) - ImX(\omega)\sin(\omega t)\, dt$$

analysis

$$ReX(\omega) = \frac{1}{\pi}\int_{-\infty}^{+\infty} x(t)\cos(\omega t)\, dt$$

$$ImX(\omega) = \frac{-1}{\pi}\int_{-\infty}^{+\infty} x(t)\sin(\omega t)\, dt$$

Time domain:
x(t) is real, continuous, and aperiodic
t runs from negative to positive infinity

Frequency domain:
Re X[ω] is real, continuous and aperiodic
Im X[ω] is real, continuous and aperiodic
ω runs from zero to positive infinity

TABLE 31-1 The Fourier Transforms